Module III — Semiconductor properties

Theory notes: concepts, results and what every symbol means

Intrinsic & extrinsic

IN ONE LINE A semiconductor is an insulator with a **small** band gap (about 1 eV). Heat breaks a few bonds and makes electron–hole pairs (intrinsic). Adding a trace of Group V or Group III atoms adds electrons or holes on purpose (extrinsic), and that is how every device is made.

What a semiconductor is

Put the three kinds of solid side by side by how well they conduct. Conductors have resistivities of 10^{-8} –$10^{-6}\ \Omega$ m, insulators 10^{7}–$10^{8}\ \Omega$ m and more, and semiconductors sit in between: 10^{-5} to $10^{4}\ \Omega$ m. Ge and Si (Group IV) have resistivities of about 0.6 and $1.5 \times 10^{3}\ \Omega$ m. Compound semiconductors come from Groups III–V (GaAs, InP) and II–VI (CdS).

The band picture from Module II explains the middle position: the valence band is full and the conduction band empty, exactly as in an insulator, but the gap E_g between them is small enough for room-temperature heat to push a few electrons across.

Group IV element	E_g at 0 K	At room temperature
C (diamond)	5.51 eV	insulator
Si	1.16 eV	semiconductor
Ge	0.75 eV	semiconductor
Sn (grey)	0.08 eV	conductor
Pb	0	conductor

Same column of the periodic table, same diamond-type bonding, and the gap shrinks steadily down the group. That single number decides insulator, semiconductor or conductor.

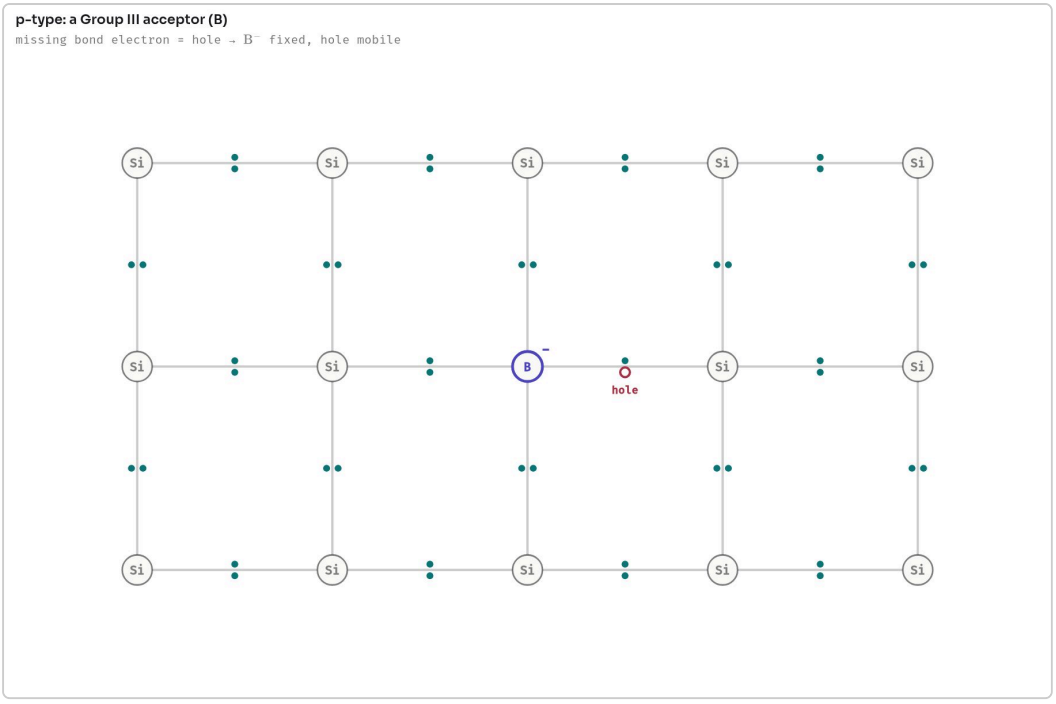
Two properties every exam asks for:

- **Negative temperature coefficient of resistance.** A metal conducts worse when hot (more scattering). A semiconductor conducts *better* when hot, because heating makes new carriers faster than it slows the old ones.
- **Doping changes everything.** Adding impurity atoms in a controlled way, about 1 in 10^{6} to 10^{8}, changes the conductivity enormously. The impurity takes the place of a host atom, so the crystal structure does not change.

The small gap also fixes the wavelength of light the material emits or absorbs: that is why semiconductors make LEDs, lasers and solar cells.

Intrinsic semiconductors

In pure Si each atom shares its four valence electrons in four covalent bonds pointing to the corners of a tetrahedron (the diamond structure of Module I). Step through the bond picture:



The bond picture, the usual way to draw it. Two dots on a link are the two electrons of one covalent bond. Step 2 is the whole of intrinsic conduction; steps 4 and 5 are doping.

At 0 K every electron is in a bond: **pure Si is an insulator at 0 K**. The valence band is full and the conduction band empty. Above 0 K a few electrons get enough thermal energy to break free. Each one leaves a **hole**, an empty place in a bond, which also carries current: a neighbouring bond electron can jump into it, so the hole moves the other way. One broken bond gives one **electron–hole pair**, so in a pure sample

$$n = p = n_i \quad \text{(intrinsic carrier concentration)}$$

WHERE

- n electrons per unit volume in the conduction band
- p holes per unit volume in the valence band
- n_i intrinsic carrier concentration of the pure crystal at that temperature

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Properties of a hole: charge $+e$; it moves in the valence band, in the direction of the field; it is not a particle but the absence of an electron in an otherwise full band; it has a **larger effective mass and smaller mobility** than an electron, because it moves in a nearly full band under the applied field *and* the ionic field of the lattice.

Generation and recombination. Pairs are made all the time by heat. A free electron that meets a hole drops back into the bond, giving out about E_g (often as light). In equilibrium the two rates balance:

$$g = R = Cn_i p_i = Cn_i^2$$

WHERE

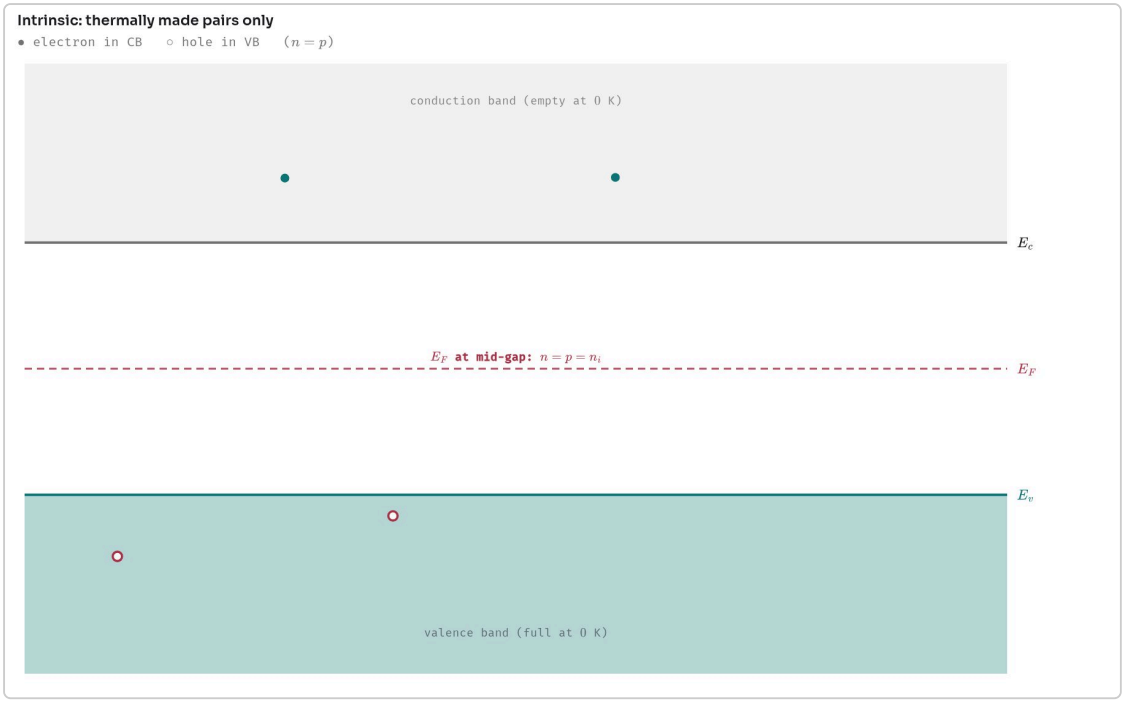
- g generation rate: electron–hole pairs made per unit volume per second
- R recombination rate: pairs lost per unit volume per second
- C constant that depends on the material
- $n_i,\ p_i$ electron and hole concentrations of the pure crystal (equal)

where C depends on the material (it is related to the densities of states at the band edges). Because the recombination rate is proportional to the product of the two concentrations, this balance is the first hint of the law of mass action, $np = n_i^2$.

Extrinsic semiconductors

n-type (donors). A Group V atom (P, As, Sb) uses four of its five valence electrons in bonds. The fifth is held only weakly, so a little energy frees it. Its energy level, the **donor level** E_d, sits in the gap just below E_c: about **0.01 eV** below in Ge and **0.03 eV** in Si. Against $kT \approx 0.026$ eV at room temperature, almost every donor gives its electron to the conduction band and becomes an immobile positive ion. Electrons are the **majority carriers**, thermally made holes the minority carriers.

p-type (acceptors). A Group III atom (B, Al, Ga, In) completes only three bonds. It takes an electron from a neighbouring Si atom to complete the fourth, leaving a hole there. Its **acceptor level** E_a is just above E_v, about 0.01 eV. The impurity becomes an immobile negative ion; holes are the majority carriers.



The same three cases as band diagrams. Switch between them: the dopant levels sit a hair inside the gap, they are almost all ionised at room temperature, and the Fermi level follows the majority carrier towards its band.

How big is the effect? Si has 5×10^{28} atoms per m^3. Doping at 1 in 10^8 gives $N_d = 5 \times 10^{20}\ \mathrm{m}^{-3}$ donors, against $n_i = 1.5 \times 10^{16}\ \mathrm{m}^{-3}$ at 300 K: some 30 000 times more electrons than the pure crystal has, from one foreign atom in a hundred million.

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In both types the **law of mass action** holds: $np = n_i^2$ at a given temperature. Doping raises one carrier and lowers the other in proportion. The crystal stays **neutral**: $n + N_a^- = p + N_d^+$.

WHERE MARKS GET LOST

A donor atom does not make a hole when it gives its electron away: what it leaves is a *fixed* positive ion, not a mobile hole. Only broken Si–Si bonds make holes. Likewise an acceptor makes a mobile hole and a fixed negative ion, not a free electron.

Drift, mobility, σ

IN ONE LINE A field gives the carriers a small extra velocity on top of their random motion: $v_d = \mu E$. Electrons and holes both carry current, so $\sigma = e(n\mu_n + p\mu_p)$. In a pure sample σ rises as $e^{-E_g/2kT}$, and $\ln\sigma$ against $1/T$ measures the gap.

Drift velocity and mobility

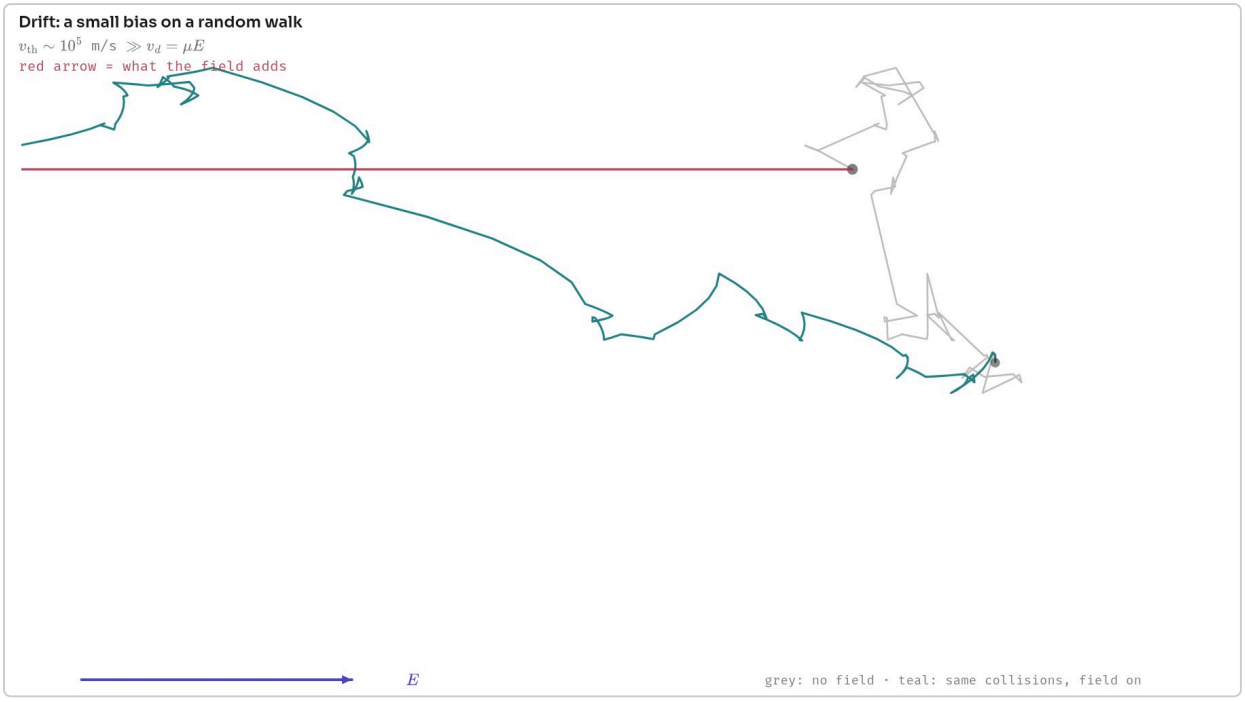
With no field the carriers move at random and give no current: for every electron going left there is one going right. A field accelerates electrons one way and holes the other, but they keep colliding (with vibrating atoms, impurities, defects) and lose the extra velocity each time. On average they gain a steady extra velocity, the **drift velocity**, proportional to the field:

$$v_d = \mu E$$

WHERE

- v_d drift velocity: the average velocity gained from the field
- μ mobility: drift velocity per unit field
- E applied electric field

The constant μ is the **mobility**: drift velocity per unit field, in $\mathrm{m^2 V^{-1} s^{-1}}$. It is the same idea as the Drude picture of Module II, with $\mu = e\tau/m^*$: a long time between collisions and a light effective mass make a mobile carrier.



One electron, the same random collisions twice: grey with no field, teal with a field. The thermal speed ($\sim 10^5$ m/s) is enormous next to the drift ($\sim$ cm/s); the field only biases the walk. The red arrow is everything the field achieved. (The bias is exaggerated a great deal so you can see it.)

Why electrons beat holes. A hole's motion is really a relay of bond electrons in a nearly full band, so it has a larger effective mass and collides more effectively: $\mu_p < \mu_n$ (Si: 0.048 against 0.135). That is why devices that use electrons (n-type channels) are faster and preferred.

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$$v_d = \mu E, \qquad J = \sigma E, \qquad \sigma = e(n\mu_n + p\mu_p)$$

WHERE

v_d	drift velocity
μ	mobility: drift velocity per unit field
E	electric field
J	current density
σ	conductivity
n, p	electron and hole concentrations
μ_n, μ_p	electron and hole mobilities
e	magnitude of the electron charge

Intrinsic: $n = p = n_i$, so $\sigma_i = e n_i(\mu_n + \mu_p)$. Electrons and holes drift opposite ways but, having opposite charges, their currents **add**.

Drift and diffusion. Carriers also move down a concentration gradient (diffusion current). In a sample with uniformly spread impurities there is no gradient, so only the drift current counts here. The two are linked by the Einstein relation $D/\mu = kT/e$ (0.026 V at room temperature).

300 K	E_g (eV)	n_i ($\mathrm{m^{-3}}$)	μ_n	μ_p	σ_i (S/m)
Si	1.1	1.5×10^{16}	0.135	0.048	4.4×10^{-4}
Ge	0.7	2.4×10^{19}	0.39	0.19	2.2

Mobilities in $\mathrm{m^2 V^{-1} s^{-1}}$. Ge conducts thousands of times better than Si, almost entirely because its smaller gap gives far more carriers.

Conductivity and temperature

Taking the mobilities as constant, all the temperature dependence comes from n_i (derived in the next section): $n_i \propto T^{3/2} e^{-E_g/2kT}$. The exponential swamps the $T^{3/2}$, so a plot of $\ln\sigma$ against $1/T$ is very nearly a straight line, and its slope gives the gap. This is the standard laboratory way of measuring E_g.

Metal against semiconductor, with impurities. Impurities in a metal add scattering, so its conductivity *falls*. Impurities in a semiconductor add carriers, so its conductivity *rises*, by many orders of magnitude, even though they also scatter a little. Temperature works the same way round: a metal's σ falls as $1/T$, a semiconductor's climbs exponentially.

WHERE MARKS GET LOST

The slope of $\ln\sigma$ against $1/T$ is $-E_g/2k$, not $-E_g/k$. Forgetting the 2 halves or doubles your band gap. Mobility data are often given in $\mathrm{cm^2 V^{-1} s^{-1}}$: divide by 10^4.

n, p, E_F, n_i

IN ONE LINE Count the states in each band ($D(E)$) times the chance they are filled ($f(E)$). Because E_F sits in the gap, f becomes a simple exponential and the counts come out as $n = N_c e^{-(E_c - E_F)/kT}$ and $p = N_v e^{-(E_F - E_v)/kT}$. Multiply them and E_F drops out: $np = n_i^2$.

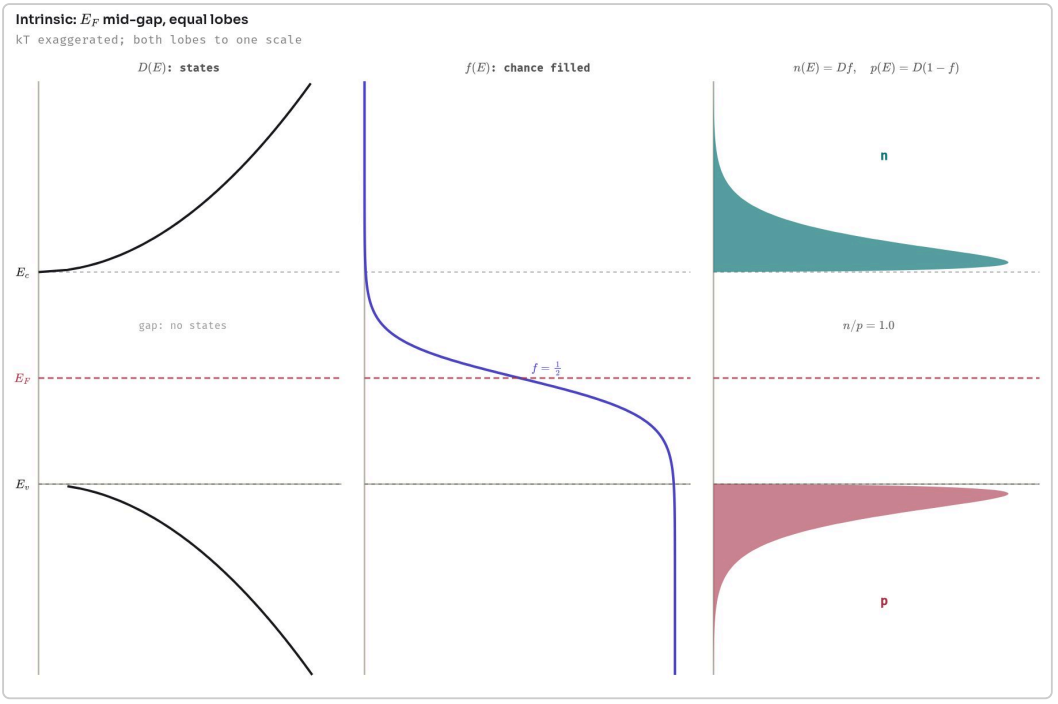
The recipe: states × occupation

Every carrier count in this module is the same two-step recipe from Module II. How many places are there at energy E (the density of states $D(E)$), and what is the chance each is occupied (the Fermi function $f(E)$)? Multiply and add up over the band:

$$n = \int_{E_c}^{\infty} D_c(E) f(E)\, dE, \qquad p = \int_{-\infty}^{E_v} D_v(E)\,[1 - f(E)] dE$$

WHERE

n	electron concentration in the conduction band
p	hole concentration in the valence band
E_c	bottom of the conduction band
E_v	top of the valence band
$D_c(E),\ D_v(E)$	density of states in each band
$f(E)$	probability that a state is filled
$1 - f(E)$	probability that a state is empty, that is, holds a hole



The three graphs drawn side by side. Left: no states in the gap, $\sqrt{E}$-shaped bands either side. Middle: the Fermi function, $\frac{1}{2}$ at E_F. Right: their product, the carriers per unit energy. Only the far tails of f reach into the bands, so the lobes hug the band edges. Move E_F with the chips and one lobe grows while the other shrinks by the same factor.

Two simplifications make the integrals easy. Near a band edge the band is parabolic, so $D(E)$ is the free-electron $\sqrt{E}$ with the effective mass. And because E_F is several kT inside the gap, $E - E_F \gg kT$ everywhere in the conduction band, so $f(E) \approx e^{-(E-E_F)/kT}$: the Fermi tail is a Maxwell–Boltzmann tail. Such a semiconductor is called **non-degenerate**.

Electrons in the conduction band

An electron at energy E in the conduction band has kinetic energy $E - E_c$. So the free-electron density of states applies with $E \to E - E_c$ and $m \to m_n^*$. Near room temperature $kT = 0.026$ eV, much smaller than $E_c - E_F$, so the Fermi function's tail is exponential.

Read the answer as "the band behaves like N_c states all sitting at E_c". N_c is the **effective density of states**: for silicon $N_c = 2.8 \times 10^{25} \left(\frac{T}{300}\right)^{3/2}\ \mathrm{m^{-3}}$, and the Boltzmann factor says what fraction of those are filled.

Holes in the valence band

A hole is a state left empty, so count $D(E)[1 - f(E)]$. Holes become exponentially rarer deeper in the valence band, the mirror image of the electrons.

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$$n = N_c e^{-(E_c - E_F)/kT}, \qquad p = N_v e^{-(E_F - E_v)/kT}$$

WHERE

$n,\ p$	electron and hole concentrations
$N_c,\ N_v$	effective density of states of the conduction and valence band
$E_c,\ E_v$	band edges
E_F	Fermi level
k	Boltzmann's constant
T	temperature

$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

WHERE

$N_c,\ N_v$	effective density of states of the two bands
$m_n^*,\ m_p^*$	effective masses of the electron and the hole
k	Boltzmann's constant
T	temperature
h	Planck's constant
2	spin degeneracy

These hold for intrinsic *and* doped samples (as long as E_F is a few kT inside the gap). For an intrinsic sample write $E_F = E_i$, the intrinsic level: $n_i = N_c e^{-(E_c - E_i)/kT}$, $p_i = N_v e^{-(E_i - E_v)/kT}$.

Where the Fermi level sits in a pure sample

In a pure sample every electron in the conduction band left a hole behind, so $n = p$. Set the two formulas equal and solve for E_F:

The second term is small: $\frac{3}{4}kT\ln(m_p^*/m_n^*)$ is a few meV for Si at room temperature, so E_F is effectively at mid-gap, rising slightly with T when holes are heavier. At 0 K it is exactly mid-gap.

Law of mass action and n_i

Multiply n by p. The E_F in the two exponents cancels, so the product depends only on the material and the temperature, never on the doping:

LEARN THIS BY HEART

$$np = n_i^2 = N_c N_v e^{-E_g/kT}, \qquad n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}\left(m_n^* m_p^*\right)^{3/4} e^{-E_g/2kT}$$

WHERE

$n,\ p$	electron and hole concentrations
n_i	intrinsic carrier concentration
$N_c,\ N_v$	effective densities of states
E_g	band gap $E_c - E_v$
$m_n^*,\ m_p^*$	effective masses
$k,\ T,\ h$	Boltzmann's constant, temperature, Planck's constant

Valid for intrinsic and extrinsic samples alike (non-degenerate).

For pure Ge at 300 K, $n_i \approx 2.4 \times 10^{19}\ \mathrm{m}^{-3}$ against 4.4×10^{28} Ge atoms per m^3: only about 5 bonds in 10^{10} are broken. Copper has about 10^{28} free electrons per m^3.

WHERE MARKS GET LOST

$n_i \propto e^{-E_g/2kT}$ but $np = n_i^2 \propto e^{-E_g/kT}$. Keep the two exponents apart. And N_c, N_v depend on temperature ($\propto T^{3/2}$): use the value at the temperature you are working at.

Extrinsic E_F & σ

IN ONE LINE In an n-type sample the electrons come from the donors, so E_F moves up, about midway between E_d and E_c. As the sample is heated E_F falls back to mid-gap and the sample behaves intrinsically. Its conductivity is $\sigma \approx N_d e \mu_n$ in the useful range.

n-type

At 0 K every donor level is full and the conduction band empty, so E_F lies between E_d and E_c. Warming ionises some donors (electrons to the conduction band) and breaks a few bonds (holes). With enough donors the holes are suppressed by mass action, so $n \approx N_d^+$.

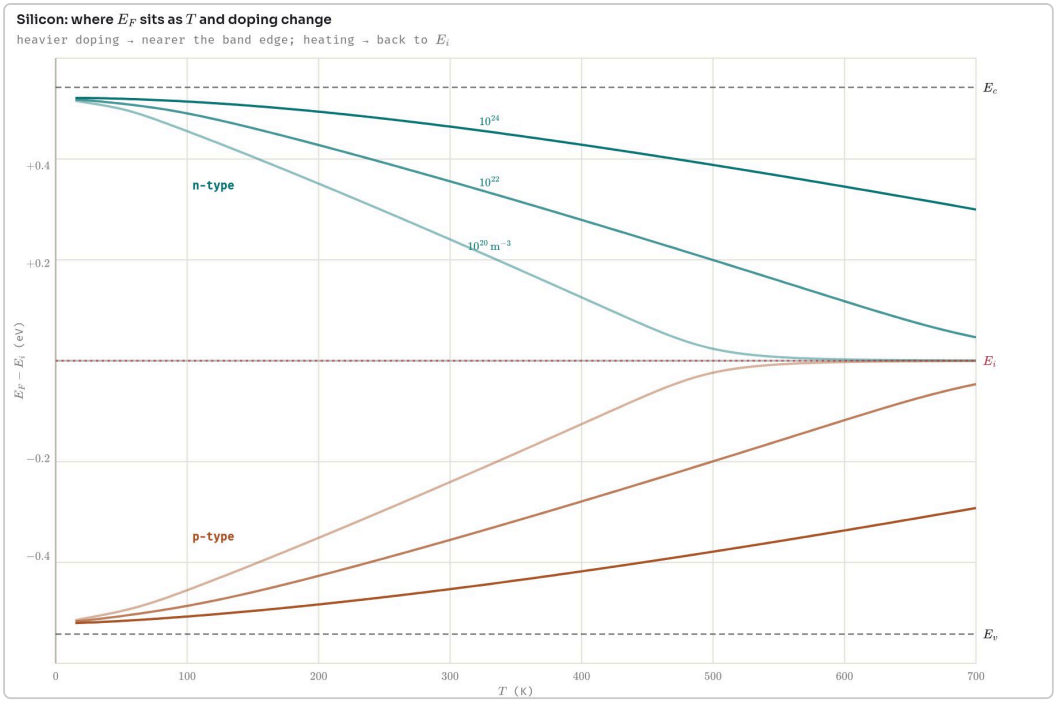
A direct link to the intrinsic level, from $n = N_c e^{-(E_c - E_F)/kT}$ and $n_i = N_c e^{-(E_c - E_i)/kT}$:

$$n = n_i e^{(E_F - E_i)/kT} \qquad \Longleftrightarrow \qquad E_F - E_i = kT \ln \frac{n}{n_i}$$

WHERE

n	electron concentration
n_i	intrinsic concentration
E_F	Fermi level of the doped sample
E_i	intrinsic Fermi level, close to mid-gap
$k,\ T$	Boltzmann's constant, temperature

So n approaches n_i as E_F approaches E_i, and grows exponentially as E_F moves towards the conduction band. Each factor of 10 in n moves E_F by $kT \ln 10 = 0.06$ eV at room temperature.



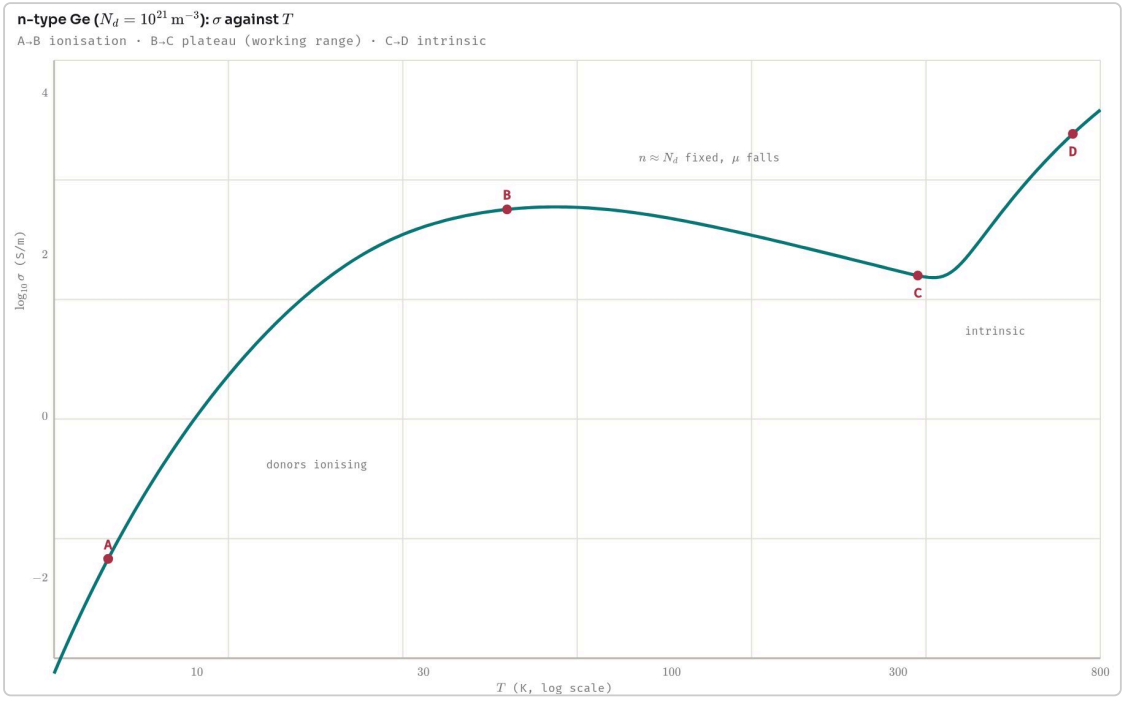
The classic curve, computed exactly from charge neutrality for silicon. Teal: n-type, copper: p-type, for 10^{20}, 10^{22} and 10^{24} impurities per m^3. Heavier doping pushes E_F towards its band edge; heating pulls every curve back to E_i as thermally made pairs swamp the dopants. The lightly doped sample turns intrinsic first.

Conductivity of n-type Ge against temperature

$$\sigma = ne\mu_n \approx eN_d\mu_n$$

WHERE

- σ conductivity of n-type material
- n electron concentration
- e electron charge
- μ_n electron mobility
- N_d donor concentration (all ionised, so $n \approx N_d$)



The classic conductivity plot for n-type Ge, computed from the same neutrality condition with a mobility that rises with T at first (impurity scattering) and then falls (lattice vibrations). The three regions are the classic 8-mark answer.

1. **A → B (about 10 K to 50 K), the ionisation region:** σ rises as more donors are ionised; maximum when all are ionised.
2. **B → C (up to about room temperature), the extrinsic plateau:** σ falls slightly: $n \approx N_d$ is fixed but the mobility drops as lattice vibrations grow. This is the working range of devices.
3. **C → D, the intrinsic region:** steep rise: broken bonds (intrinsic carriers) overtake the donors, and the curve joins the intrinsic line.

p-type

Same argument with acceptors: $p \approx N_a^- = N_a f(E_a) \approx N_a e^{-(E_a - E_F)/kT}$, and equating with $p = N_v e^{-(E_F - E_v)/kT}$:

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$$E_F = \frac{E_v + E_a}{2} - \frac{kT}{2} \ln \frac{N_a}{N_v}, \qquad p = n_i e^{(E_i - E_F)/kT}, \qquad \sigma_p = ep\mu_p \approx eN_a\mu_p$$

WHERE

E_F	Fermi level of p-type material
E_v	top of the valence band
E_a	acceptor level
N_a	acceptor concentration
N_v	effective density of states of the valence band
p	hole concentration
E_i	intrinsic Fermi level
σ_p	conductivity of p-type material
μ_p	hole mobility
k, T	Boltzmann's constant, temperature

E_F lies about midway between E_v and E_a, moves up with rising T and finally meets E_i. Not valid at very high T (then n cannot be neglected).

More impurities give higher conductivity: for 10^{20}–10^{22} impurities per m^3 the resistivity of Si and Ge falls to 10^{-3}–$10^{-1}\ \Omega\,\mathrm{m}$. Very heavy doping ($10^{23}$–$10^{24}\ \mathrm{m}^{-3}$) makes **degenerate semiconductors**: E_F moves into the band itself, the Boltzmann approximation fails, and the sample conducts nearly as well as a metal (high-power and high-frequency devices).

Mixed semiconductors

With both donors and acceptors, neutrality reads $N_d^+ + p = N_a^- + n$. If $N_d = N_a$, the acceptors capture the donors' electrons: $n = p = n_i$, and the sample behaves as intrinsic (**compensated**). If $N_d \neq N_a$ it is n-type or p-type according to whichever is larger, with an effective doping of $|N_d - N_a|$.

LEARN THIS BY HEART

Summary of the Fermi level: intrinsic, at mid-gap; n-type, about midway between E_d and E_c; p-type, about midway between E_v and E_a. As T rises it approaches the intrinsic level in every case.